OpenPipeline Migration Guide: Part 3

> **Series:** OPMIG | **Notebook:** 3 of 9 | **Created:** December 2025 | **Last Updated:** 04/04/2026

Migration Assessment & Planning

Table of Contents

Learning Objectives

By the end of this notebook, you will:

- ✅ Discover and analyze your current data landscape

- ✅ **Use Davis Copilot for intelligent discovery**
- ✅ Assess parsing requirements and coverage
- ✅ Identify cost optimization opportunities
- ✅ Detect sensitive data requiring masking
- ✅ **Prioritize sources using scoring matrix**
- ✅ **Create data-driven migration waves**
- ✅ Generate comprehensive source inventory

Discovery: Current Data Landscape

Start by understanding what data you're currently ingesting and how it's being processed.

Using Davis Copilot for Discovery

If you have access to the Dynatrace MCP Server, you can use Davis Copilot to assist with migration discovery.

Davis Copilot Capabilities

Tool	Purpose	Example Use Case
chat_with_davis_copilot	Ask any Dynatrace question	"What log sources exist in my environment?"
generate_dql_from_natural_language	Natural language → DQL	"Show me error logs from the last week"
explain_dql_in_natural_language	Explain complex queries	Understand existing queries

Tool	Purpose	Example Use Case
verify_dql	Validate DQL syntax	Check queries before running
execute_dql	Run queries against tenant	Get real data

Example: Discovery with Davis Copilot

Question to Davis:

```
"What are the top 10 log sources by volume in the last 7 days?"
```

Davis generates DQL:

```
fetch logs, from: now() - 7d
| summarize {log_count = count()}, by: {log.source}
| sort log_count desc
| limit 10
```

Question to Davis:

```
"Show me logs that might contain credit card numbers"
```

Davis generates DQL:

```
// Security pattern detection - look for potential credit card or payment
data
fetch logs, from: now() - 1h
| filter contains(toString(content), "card") OR contains(toString(content),
"payment") OR contains(toString(content), "credit")
| summarize {count = count()}, by: {log.source}
| sort count desc
```

Benefits of MCP-Assisted Discovery

Benefit	Description
Natural Language	No need to know DQL syntax
Real Data	Execute against live tenant

Benefit	Description
Syntax Validation	Catch errors before execution
Explanation	Understand complex patterns
Faster Discovery	Iterate quickly on queries

> 💡 **Tip:** Use Davis Copilot iteratively. Ask initial questions, review results, then refine your questions based on what you discover.

Discovery Workflow with MCP

```
1. ASK DAVIS
   └ "What log sources exist?"

2. DAVIS GENERATES DQL
   └ fetch logs | summarize...

3. VERIFY SYNTAX
   └ mcp verify_dql(query)

4. EXECUTE
   └ mcp execute_dql(query)

5. ANALYZE RESULTS
   └ Identify patterns, volumes

6. REFINE
   └ Ask follow-up questions
```

```
// Overall data volume summary by data type
// Understand the scale of your migration
fetch logs, from: now() - 7d
| summarize {log_count = count()}
| fieldsAdd data_type = "logs", daily_avg = log_count / 7
```

```
// Compare logs processed by OpenPipeline vs potentially classic
// Identifies migration progress
fetch logs, from: now() - 7d
| fieldsAdd processing_type = if(isNotNull(dt.openpipeline.source),
"OpenPipeline", else: "Unknown/Classic")
| summarize {record_count = count()}, by: {processing_type}
| fieldsAdd daily_avg = record_count / 7
| sort record_count desc
```

```
// Breakdown by OpenPipeline source type
// Shows which ingestion methods are being used
fetch logs, from: now() - 7d
| summarize {record_count = count()}, by: {dt.openpipeline.source}
| fieldsAdd daily_avg = round(record_count / 7, decimals: 0)
| sort record_count desc
```

```
// Check span ingestion landscape (OpenPipeline handles spans too!)
fetch spans, from: now() - 7d
| summarize {span_count = count()}, by: {dt.openpipeline.source}
| fieldsAdd daily_avg = round(span_count / 7, decimals: 0)
| sort span_count desc
```

Volume Analysis

Understanding volume patterns helps you prioritize migration and plan for cost optimization.

```
// Daily log volume trend
// Identify growth patterns and peak days
fetch logs, from: now() - 30d
| makeTimeseries {daily_count = count()}, interval: 1d
```

```
// Hourly volume pattern (typical day)
// Understand when peak ingestion occurs
fetch logs, from: now() - 7d
| fieldsAdd hour_of_day = getHour(timestamp)
| summarize {avg_hourly = count() / 7}, by: {hour_of_day}
| sort hour_of_day asc
```

```
// Volume by log source - TOP 25
// Identify highest volume sources for prioritization
fetch logs, from: now() - 7d
| summarize {record_count = count()}, by: {log.source}
| sort record_count desc
| limit 25
| fieldsAdd daily_avg = round(record_count / 7, decimals: 0)
```

```
// Volume by host entity
// Identify which hosts generate the most logs
fetch logs, from: now() - 7d
| filter isNotNull(dt.entity.host)
| summarize {record_count = count()}, by: {dt.entity.host}
| sort record_count desc
| limit 20
| fieldsAdd daily_avg = round(record_count / 7, decimals: 0)
```

```
// Volume by Kubernetes namespace (if applicable)
fetch logs, from: now() - 7d
| filter isNotNull(k8s.namespace.name)
| summarize {record_count = count()}, by: {k8s.namespace.name}
| sort record_count desc
| limit 20
| fieldsAdd daily_avg = round(record_count / 7, decimals: 0)
```

Source Inventory

Create a comprehensive inventory of log sources for migration planning.

```
// Complete log source inventory with characteristics
fetch logs, from: now() - 7d
| summarize {
    total_records = count(),
    unique_hosts = countDistinct(dt.entity.host),
    has_loglevel = countIf(isNotNull(loglevel)),
    error_count = countIf(loglevel == "ERROR" OR status == "ERROR")
  }, by: {log.source}
| fieldsAdd parsing_coverage = round((toDouble(has_loglevel) /
toDouble(total_records)) * 100, decimals: 1)
| fieldsAdd error_rate = round((toDouble(error_count) /
toDouble(total_records)) * 100, decimals: 2)
| sort total_records desc
| limit 30
```

```
// Log sources by ingestion method
// Map sources to their OpenPipeline ingestion type
fetch logs, from: now() - 7d
| summarize {record_count = count()}, by: {log.source,
dt.openpipeline.source}
| sort record_count desc
| limit 50
```

```
// Log sources by current pipeline assignment
// Identify which sources already have custom pipelines
fetch logs, from: now() - 7d
| summarize {record_count = count()}, by: {log.source,
dt.openpipeline.pipelines}
| sort record_count desc
| limit 50
```

```
// Log sources by bucket
// Shows current storage distribution
fetch logs, from: now() - 7d
| summarize {record_count = count()}, by: {log.source, dt.system.bucket}
| sort record_count desc
| limit 50
```

Parsing Requirements Analysis

Identify which log sources need parsing pipelines to extract structured data.

```
// Overall parsing coverage assessment
// What percentage of logs have structured log levels?
fetch logs, from: now() - 24h
| summarize {
    total = count(),
    with_loglevel = countIf(isNotNull(loglevel)),
    with_status = countIf(isNotNull(status)),
    either = countIf(isNotNull(loglevel) OR isNotNull(status))
  }
| fieldsAdd loglevel_coverage = round((toDouble(with_loglevel) /
toDouble(total)) * 100, decimals: 1)
| fieldsAdd status_coverage = round((toDouble(with_status) / toDouble(total))
* 100, decimals: 1)
| fieldsAdd overall_coverage = round((toDouble(either) / toDouble(total)) *
100, decimals: 1)
```

```
// Sources with low parsing coverage (need custom parsing)
// These sources should be prioritized for parsing pipelines
fetch logs, from: now() - 7d
| summarize {
    total_records = count(),
    parsed = countIf(isNotNull(loglevel) OR isNotNull(status))
  }, by: {log.source}
| filter total_records &gt; 1000
| fieldsAdd parsing_pct = round((toDouble(parsed) / toDouble(total_records))
* 100, decimals: 1)
| filter parsing_pct &lt; 50
| sort total_records desc
| limit 25
```

```
// Sample unparsed log content for pattern analysis
// Review these to design parsing rules
fetch logs, from: now() - 1h
| filter isNull(loglevel) AND isNull(status)
| fields timestamp, log.source, content
| limit 50
```

```
// Identify JSON logs that could benefit from JSON parsing
// JSON logs can be automatically parsed in OpenPipeline
fetch logs, from: now() - 1h
| filter startsWith(toString(content), "{")
| summarize {json_logs = count()}, by: {log.source}
| sort json_logs desc
| limit 20
```

```
// Sample JSON logs for structure analysis
fetch logs, from: now() - 1h
| filter startsWith(toString(content), "{")
| fields timestamp, log.source, content
| limit 20
```

Cost Optimization Opportunities

Identify logs that can be dropped or routed to shorter retention buckets to reduce costs.

```
// Debug log volume - prime candidates for dropping
fetch logs, from: now() - 7d
| summarize {
    total = count(),
    debug_logs = countIf(loglevel == "DEBUG" OR status == "DEBUG"),
    trace_logs = countIf(loglevel == "TRACE" OR status == "TRACE")
  }
| fieldsAdd debug_pct = round((toDouble(debug_logs) / toDouble(total)) * 100,
decimals: 2)
| fieldsAdd trace_pct = round((toDouble(trace_logs) / toDouble(total)) * 100,
decimals: 2)
| fieldsAdd droppable_pct = round((toDouble(debug_logs + trace_logs) /
toDouble(total)) * 100, decimals: 2)
```

```
// Health check and metrics endpoint logs — often droppable
fetch logs, from: now() - 7d
| summarize {
    total = count(),
    health_checks = countIf(contains(toString(content), "health") OR contains(toString(content), "healthz")),
    readiness = countIf(contains(toString(content), "ready") OR contains(toString(content), "readiness")),
    liveness = countIf(contains(toString(content), "alive") OR contains(toString(content), "liveness")),
    metrics = countIf(contains(toString(content), "/metrics") OR contains(toString(content), "/prometheus"))
  }
| fieldsAdd droppable = health_checks + readiness + liveness + metrics
| fieldsAdd savings_pct = round((toDouble(droppable) / toDouble(total)) * 100, decimals: 2)
```

```
// Log level distribution by source
// Identify sources with high debug/trace output
fetch logs, from: now() - 7d
| filter isNotNull(loglevel)
| summarize {record_count = count()}, by: {log.source, loglevel}
| sort log.source asc, record_count desc
```

```
// Highly repetitive log patterns (noise candidates)
// Find logs with same content appearing many times
fetch logs, from: now() - 1h
| summarize {occurrences = count()}, by: {content}
| filter occurrences &gt; 100
| sort occurrences desc
| limit 25
```

```
// Estimate potential cost savings from dropping debug/noise
fetch logs, from: now() - 7d
| summarize {
    total = count(),
    droppable_debug = countIf(loglevel == "DEBUG" OR status == "DEBUG" OR
loglevel == "TRACE"),
    droppable_health = countIf(contains(toString(content), "health") OR
contains(toString(content), "/metrics")),
    info_logs = countIf(loglevel == "INFO" OR status == "INFO")
  }, by: {log.source}
| fieldsAdd potential_drops = droppable_debug + droppable_health
| fieldsAdd drop_pct = round((toDouble(potential_drops) / toDouble(total)) *
100, decimals: 1)
| filter potential_drops &gt; 100
| sort potential_drops desc
| limit 25
```

Migration Priority Scoring Matrix

Use this data-driven approach to prioritize which log sources to migrate first.

Scoring Factors

Factor	Weight	High Score (5)	Low Score (1)
Volume	25%	>1M logs/day	<10K logs/day
Cost Savings Potential	25%	>50% droppable	<5% droppable
Security Risk	25%	Contains PII/PHI	No sensitive data
Parsing Complexity	10%	JSON/simple format	Complex multi-format
Business Criticality	15%	Production core services	Dev/test environments

Calculating Priority Score

```
Priority Score =
  (Volume Score × 0.25) +
  (Cost Savings Score × 0.25) +
  (Security Risk Score × 0.25) +
  (Parsing Complexity Score × 0.10) +
  (Business Criticality Score × 0.15)

Range: 1.0 (lowest) to 5.0 (highest)
```

Priority Tiers

Score Range	Tier	Action
4.0 - 5.0	🔴 Critical	Wave 1 - Migrate immediately
3.0 - 3.9	🟠 High	Wave 2 - Migrate within 2 weeks
2.0 - 2.9	🟡 Medium	Wave 3 - Migrate within 1 month
1.0 - 1.9	🟢 Low	Wave 4 - Migrate when convenient

Example: E-Commerce Logs

Source: `payment-service`

Factor	Score	Justification
Volume	5	2M logs/day
Cost Savings	4	60% debug logs droppable
Security Risk	5	Contains credit card numbers
Parsing	4	JSON format (easy)
Criticality	5	Core payment processing

Priority Score:

```
(5 × 0.25) + (4 × 0.25) + (5 × 0.25) + (4 × 0.10) + (5 × 0.15) = 4.65
```

Result: 🔴 Wave 1 - Critical Priority

Example: Development Logs

Source: `dev-test-service`

Factor	Score	Justification
Volume	2	50K logs/day
Cost Savings	3	30% debug logs
Security Risk	1	No sensitive data
Parsing	3	Moderate complexity
Criticality	1	Non-production

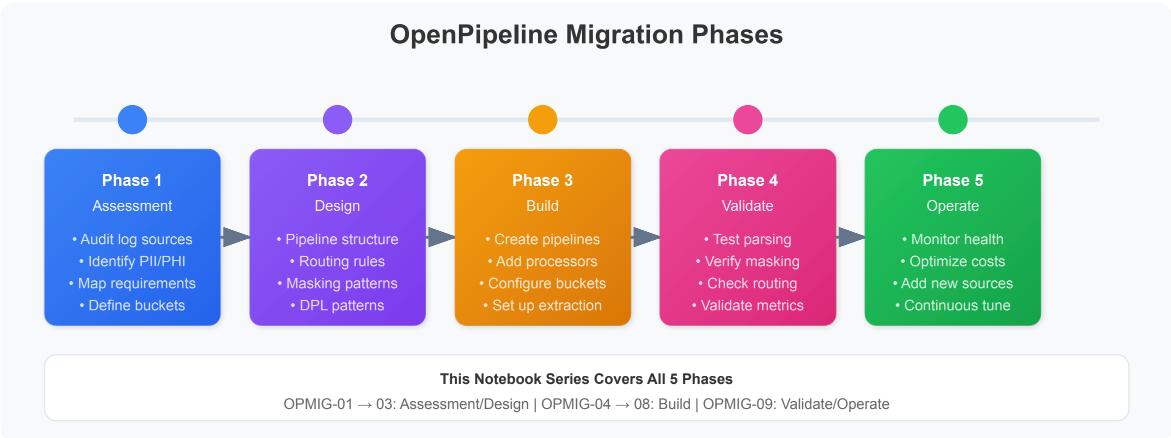
Priority Score:

```
(2 × 0.25) + (3 × 0.25) + (1 × 0.25) + (3 × 0.10) + (1 × 0.15) = 1.95
```

Result: 🟢 **Wave 4 - Low Priority**

Defining Migration Waves

Organize your migration into manageable waves based on priority scores.



Wave 1: Critical (Weeks 1-2)

Criteria:
- Priority score ≥ 4.0
- High security risk OR high business criticality
- Quick wins with significant impact

Typical Sources:
- Payment/financial transaction logs
- Authentication/authorization logs
- Production API gateways
- Customer-facing services

Migration Activities:
1. Configure masking for sensitive data
2. Set up high-retention buckets (90+ days)
3. Extract critical business metrics
4. Implement drop rules for noise
5. Validate thoroughly before cutover

Wave 2: High Priority (Weeks 3-4)

Criteria:
- Priority score 3.0 - 3.9
- High volume sources
- Significant cost savings potential

Typical Sources:
- High-volume application logs
- Microservices with debug logs
- Infrastructure logs (K8s, cloud)
- Database audit logs

Migration Activities:
1. Implement tiered bucket strategy
2. Configure aggressive drop rules
3. Extract SLI/SLO metrics
4. Parse for standard fields

Wave 3: Medium Priority (Weeks 5-6)

Criteria:
- Priority score 2.0 - 2.9
- Moderate volume
- Standard processing requirements

Typical Sources:
- Supporting microservices
- Batch processing logs
- Background workers
- Scheduled jobs

Migration Activities:
1. Use default bucket with standard retention
2. Basic parsing and enrichment
3. Standard drop rules

Wave 4: Low Priority (Weeks 7+)

Criteria:
- Priority score < 2.0
- Low volume or non-critical
- Development/test environments

Typical Sources:
- Development environment logs
- Test automation logs
- Experimental services
- Legacy systems (soon to be retired)

Migration Activities:
1. Short retention buckets (7 days)
2. Minimal processing
3. May consider NOT migrating if retiring soon

Wave Planning Template

Wave	Timeline	Sources	Expected Outcome
Wave 1	Week 1-2	5-10 critical sources	Security compliance, audit logs

Wave	Timeline	Sources	Expected Outcome
Wave 2	Week 3-4	10-20 high-volume sources	50%+ cost savings
Wave 3	Week 5-6	20-30 standard sources	Complete production migration
Wave 4	Week 7+	Remaining sources	100% migration complete

Risk Mitigation Per Wave

Wave	Mitigation Strategy
Wave 1	Extra validation, staged rollout, 24/7 monitoring
Wave 2	Standard validation, rollout groups, business hours monitoring
Wave 3	Standard process, batch rollout
Wave 4	Minimal ceremony, quick rollout

```
// Generate priority scoring data for your log sources
// Use this data to calculate priority scores manually

fetch logs, from: now() - 7d
| summarize {
    // VOLUME FACTOR
    total_records = count(),
    daily_avg = count() / 7,

    // COST SAVINGS FACTOR
    debug_count = countIf(loglevel == "DEBUG" or status == "DEBUG"),
    trace_count = countIf(loglevel == "TRACE" or status == "TRACE"),
    health_checks = countIf(contains(toString(content), "health") or
contains(toString(content), "/ready")),

    // SECURITY FACTOR (indicators)
    potential_pii = countIf(
        contains(toString(content), "@") OR
        contains(toString(content), "card") OR
        contains(toString(content), "ssn") OR
        contains(toString(content), "password")
    ),

    // PARSING FACTOR
    parsed_records = countIf(isNotNull(loglevel)),

    // ERROR RATE (criticality indicator)
    error_count = countIf(loglevel == "ERROR" or status == "ERROR"),

    unique_hosts = countDistinct(dt.entity.host)
  }, by: {log.source}
| filter total_records &gt; 1000  // Only sources with meaningful volume
| fieldsAdd droppable = debug_count + trace_count + health_checks
| fieldsAdd drop_pct = round((toDouble(droppable) / toDouble(total_records))
* 100, decimals: 1)
| fieldsAdd parsing_pct = round((toDouble(parsed_records) /
toDouble(total_records)) * 100, decimals: 1)
| fieldsAdd error_rate = round((toDouble(error_count) /
toDouble(total_records)) * 100, decimals: 2)
| fieldsAdd pii_pct = round((toDouble(potential_pii) /
toDouble(total_records)) * 100, decimals: 2)
| sort total_records desc
| limit 50
```

Security & Compliance Assessment

Identify logs containing sensitive data that needs masking before storage.

```
// Search for potential PII patterns
// CAUTION: Review results carefully for actual sensitive data
fetch logs, from: now() - 1h
| filter contains(toString(content), "email")
    OR contains(toString(content), "password")
    OR contains(toString(content), "credit")
    OR contains(toString(content), "ssn")
    OR contains(toString(content), "social security")
| summarize {potential_pii = count()}, by: {log.source}
| sort potential_pii desc
| limit 20
```

```
// Look for email patterns in logs
fetch logs, from: now() - 1h
| filter contains(toString(content), "@")
| summarize {email_patterns = count()}, by: {log.source}
| sort email_patterns desc
| limit 20
```

```
// Sample logs with potential sensitive data for review
// Review these to design masking rules
fetch logs, from: now() - 1h
| filter contains(toString(content), "email") OR contains(toString(content), "password")
| fields timestamp, log.source, content
| limit 25
```

```
// Look for IP addresses in logs (may need masking for GDPR)
fetch logs, from: now() - 1h
| filter contains(toString(content), ".")
| summarize {logs_with_dots = count()}, by: {log.source}
| sort logs_with_dots desc
| limit 20
```

```
// Audit log sources - may need longer retention
fetch logs, from: now() - 7d
| filter contains(toString(log.source), "audit")
    OR contains(toString(content), "audit")
    OR contains(toString(content), "login")
    OR contains(toString(content), "authentication")
| summarize {audit_logs = count()}, by: {log.source}
| sort audit_logs desc
| limit 20
```

Migration Priority Matrix

Based on your assessment, prioritize sources for migration using this framework:

Priority Scoring Criteria

Factor	Weight	High Score	Low Score
Volume	30%	>1M logs/day	<10K logs/day
Cost Savings Potential	25%	>30% droppable	<5% droppable
Security Risk	25%	Contains PII	No sensitive data
Parsing Complexity	10%	Simple JSON/structured	Complex multi-format
Business Criticality	10%	Core business app	Development/test

Recommended Migration Waves

Wave	Priority	Criteria	Examples
Wave 1	🔴 Critical	High volume + security risk	Payment logs, auth logs
Wave 2	🟠 High	High volume + cost savings	Debug-heavy apps, health checks
Wave 3	🟡 Medium	Medium volume, standard processing	Application logs

Wave	Priority	Criteria	Examples
Wave 4	🟢 Low	Low volume, minimal processing	Development, test environments

```
// Generate migration priority assessment by source
fetch logs, from: now() - 7d
| summarize {
    total_records = count(),
    debug_count = countIf(loglevel == "DEBUG" OR status == "DEBUG"),
    error_count = countIf(loglevel == "ERROR" OR status == "ERROR"),
    has_parsing = countIf(isNotNull(loglevel)),
    unique_hosts = countDistinct(dt.entity.host)
  }, by: {log.source}
| filter total_records &gt; 1000
| fieldsAdd daily_avg = round(total_records / 7, decimals: 0)
| fieldsAdd drop_opportunity = round((toDouble(debug_count) /
toDouble(total_records)) * 100, decimals: 1)
| fieldsAdd parsing_coverage = round((toDouble(has_parsing) /
toDouble(total_records)) * 100, decimals: 1)
| fieldsAdd error_rate = round((toDouble(error_count) /
toDouble(total_records)) * 100, decimals: 2)
| sort total_records desc
| limit 30
```

Planning Your Migration

Migration Checklist

Use this checklist to track your migration planning:

Discovery Phase ✓

- [ ] Inventoried all log sources
- [ ] Documented volume by source
- [ ] Identified OpenPipeline sources vs classic
- [ ] Analyzed current bucket distribution

Assessment Phase ✓

- [ ] Identified parsing requirements
- [ ] Calculated cost savings opportunities
- [ ] Found sources with sensitive data
- [ ] Assigned migration priorities

Planning Phase

- [ ] Defined migration waves
- [ ] Designed pipeline structure
- [ ] Planned routing rules
- [ ] Documented parsing patterns (DPL)
- [ ] Defined masking requirements
- [ ] Planned bucket strategy

Pre-Migration Testing

- [ ] Tested parsing patterns in DPL Architect
- [ ] Validated masking rules
- [ ] Tested routing conditions
- [ ] Verified metric extraction

Pipeline Design Template

For each source being migrated, document:

Aspect	Details
Source Name	(e.g., `nginx-access`)
Volume	(e.g., `500K/day`)
Pipeline Name	(e.g., `nginx-logs`)
Routing Condition	(e.g., `log.source == "nginx"`)
Parsing Required?	Yes/No
Parse Pattern	(DPL pattern)

Aspect	Details
Masking Required?	Yes/No
Masking Rules	(fields to mask)
Drop Rules	(conditions to drop)
Metric Extraction	(metrics to create)
Target Bucket	(bucket name)
Retention	(days)

Assessment Summary Export

Run these queries to create a summary you can export and share with your team.

```
// Complete source inventory with all metrics
// Export this for migration planning documentation
fetch logs, from: now() - 7d
| summarize {
    total_records = count(),
    daily_avg = count() / 7,
    unique_hosts = countDistinct(dt.entity.host),
    debug_count = countIf(loglevel == "DEBUG" OR status == "DEBUG"),
    error_count = countIf(loglevel == "ERROR" OR status == "ERROR"),
    parsed_count = countIf(isNotNull(loglevel))
  }, by: {log.source, dt.openpipeline.source, dt.system.bucket}
| fieldsAdd parsing_pct = round((toDouble(parsed_count) /
toDouble(total_records)) * 100, decimals: 1)
| fieldsAdd drop_opportunity_pct = round((toDouble(debug_count) /
toDouble(total_records)) * 100, decimals: 1)
| fieldsAdd error_rate_pct = round((toDouble(error_count) /
toDouble(total_records)) * 100, decimals: 2)
| sort total_records desc
| limit 100
```

Next Steps

With your assessment complete, continue with:

Notebook	Focus Area
OPMIG-04	Pipeline Configuration Fundamentals
OPMIG-05	Routing & Bucket Management
OPMIG-06	Processing, Parsing & Transformation
OPMIG-07	Metric & Event Extraction
OPMIG-08	Security, Masking & Compliance

References

- OpenPipeline Configuration Tutorial
- Log Management Limits
- Grail Buckets

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